Alumni Networking Portal

with Job Posting and Communication Features

Not just a portal — an institutional Alumni platform

Avanthi Institute of Engineering and Technology, Makavarapalem

Presented By:

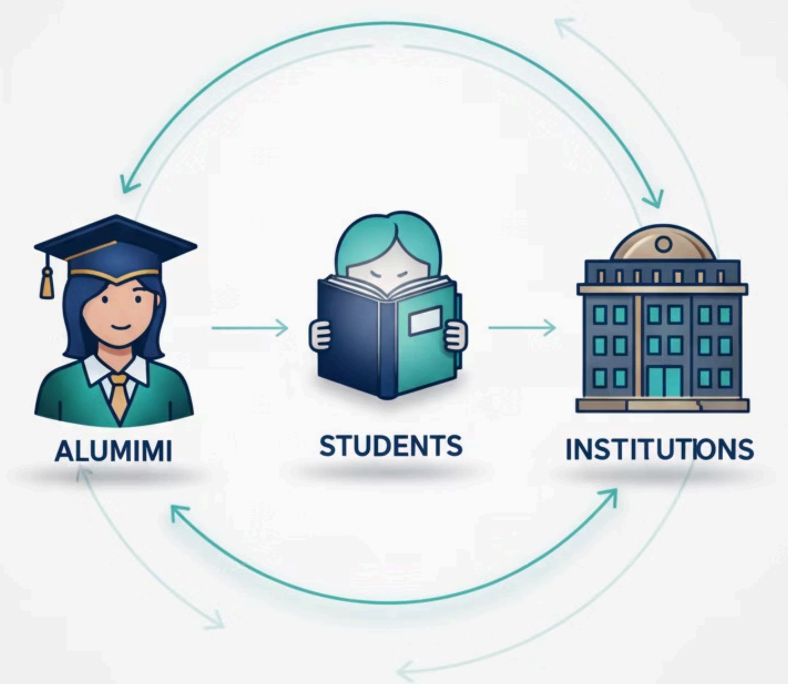
- M. Bhargavi (22811A05A1)
- K. Murali Karthik (22811A0583)
- P. Maha Lakshmi (22811A05B7)
- K. Chinni Krishna (22811A0576)
- M. Pranay Kumar (23815A0519)
- L. Rama Krishna (23815A0515)

Project Guide:

Ms. Dhana Lakshmi
Assistant Professor
Department of CSE

Presentation Overview

Introduction	Project Abstract
Existing System	Proposed System
Literature Review	System Architecture
Software Requirements Specification	Modules
Data Flow Diagram	UML Diagram
Conclusion	References



PROJECT OVERVIEW

What is the Project?

A centralized and secure alumni networking platform that connects verified alumni, students, and institutions for job opportunities, mentorship and professional communication — powered by AI-driven insights.

- **Alumni Share Opportunities:** Post job openings, mentorship offers, and announcements
- **Students Discover & Connect:** Explore opportunities, apply for jobs, and communicate with alumni
- **Institutions Gain Insights:** Access verified alumni data and career analytics

PROJECT OVERVIEW

Project Context & Scope

Modern institutions struggle with effectively engaging growing student and alumni populations, tracking career outcomes, and understanding long-term professional impact. This project provides a centralized platform to overcome these challenges, fostering sustained alumni interaction and seamless academic-industry continuity.

Project Scope Includes

- Academic and career engagement
- Centralized communication
- Career outcome visibility
- Alumni journey insights

Stakeholders Covered

- Students
- Alumni
- Faculty & Administrators

PROBLEM ANALYSIS

Existing Solutions & Their Limitations

Current tools used by students and institutions

LinkedIn : Not institution-specific, lacks verification

Email-based systems: Unstructured and difficult to track

WhatsApp Groups: Informal, unstructured, no verification

Facebook Groups: Open platform, low authenticity, poor control

Why They Fail

- Not institution-specific or verified
- Information scattered across platforms
- No structured job tracking
- Manual and time-consuming processes
- No analytics or long-term engagement

Gaps in Alumni Engagement

- Disconnected Networks
- Inefficient Job Matching
- Limited Institutional Insight
- Suboptimal Communication

OUR SOLUTION

Proposed Solution

The Alumni Networking Portal is a centralized, web-based system designed to connect verified alumni, students, and administrators for job sharing, mentorship, and communication, ensuring secure access, efficient data management, and institutional oversight.


Alumni


Students


Administrator

Central Portal

Core Functional Features

Alumni Registration & Profile Management

Secure registration with robust profile verification.

Alumni Directory with Search

Searchable database to discover alumni by skills, location, and industry

Job & Internship System

Alumni-driven opportunities with structured applications.

Secure Communication

Trusted messaging for alumni–student mentorship.

Admin Management

User approval, access control, and platform governance.

Event Announcements & Notifications

Real-time updates on networking events and opportunities

LITERATURE REVIEW

Literature Survey

A compilation of the research, platforms, and technologies that informed this project.

Research Papers

1. IRJET (2025). AlumniConnect – A Digital Bridge Between Students and Alumni. **https://www.irjet.net**
2. IEEE Xplore (2019). Online Alumni Management System Using Web Technologies. **https://ieeexplore.ieee.org**

Platforms Studied

1. LinkedIn Official Website – **https://www.linkedin.com**
2. AlmaConnect – Alumni Engagement Platform – **https://www.almaconnect.com**

Technologies & Documentation

1. OpenAI API Documentation – **https://platform.openai.com/docs**
2. Flask Documentation – **https://flask.palletsprojects.com**
3. MongoDB Documentation – **https://www.mongodb.com/docs**

System Architecture

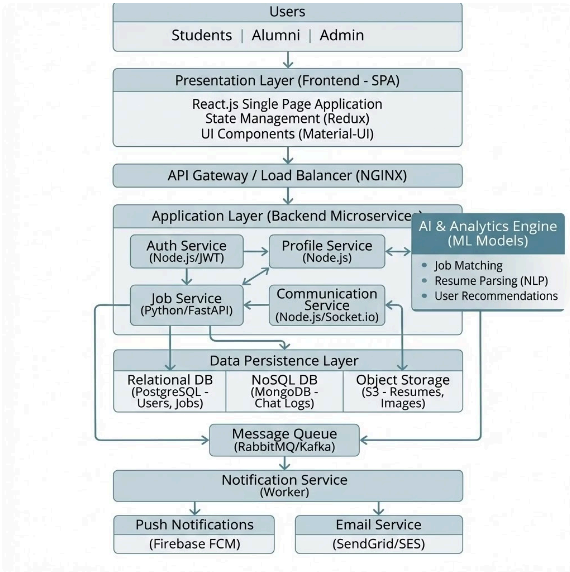
→ Frontend Layer
HTML5, CSS3, JavaScript with Bootstrap

→ Backend Services
Python (Django/Flask)

→ Database Management
MySQL, PostgreSQL, or MongoDB

→ Notification System
Firebase Cloud Messaging

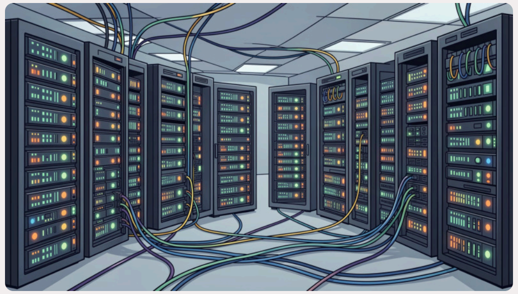
→ Development Tools
GitHub, VS Code , and CI/CD pipelines



Made with GAMMA

SOFTWARE REQUIREMENTS

Software Requirements Specification (SRS)



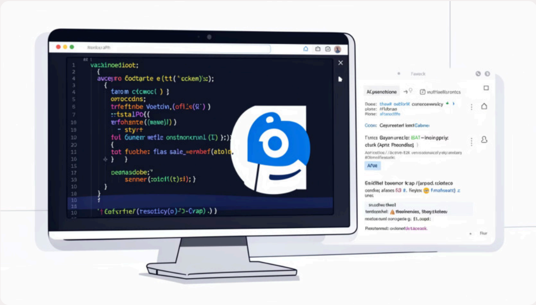
Hardware Requirements

Server :

- Intel i3 or above
- 8GB RAM minimum
- 256GB SSD

Client:

- 4GB RAM
- Modern browser (Chrome/Edge/Firefox)
- Internet connection



Software Requirements

- Frontend: HTML, CSS, JavaScript, React
- Backend: Flask / Node.js
- Database: MySQL / MongoDB
- Tools: VS Code, GitHub

ARCHITECTURE OVERVIEW

System Modules

User Management & Authentication Module

Secure, role-based login system categorizing users into Students, Alumni, and Administrators.

Profile & Admin Verification Module

Ensures network authenticity. Alumni and students must be explicitly verified by the Admin before they can view directories or post jobs.

AI-Powered Job Board Module

Alumni can post jobs while the system's background Machine Learning service extracts data from uploaded student PDF resumes to generate semantic embeddings and calculate job match scores.

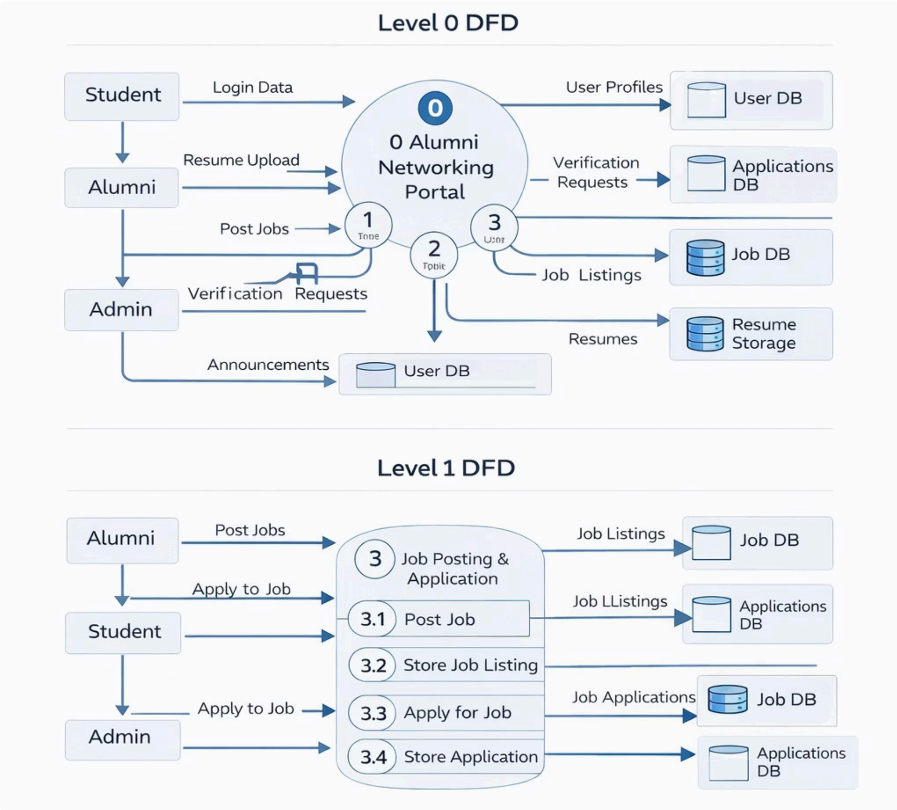
Secure Communication Module

Features a built-in one-on-one messaging system for direct mentorship, alongside a global announcement board for events.

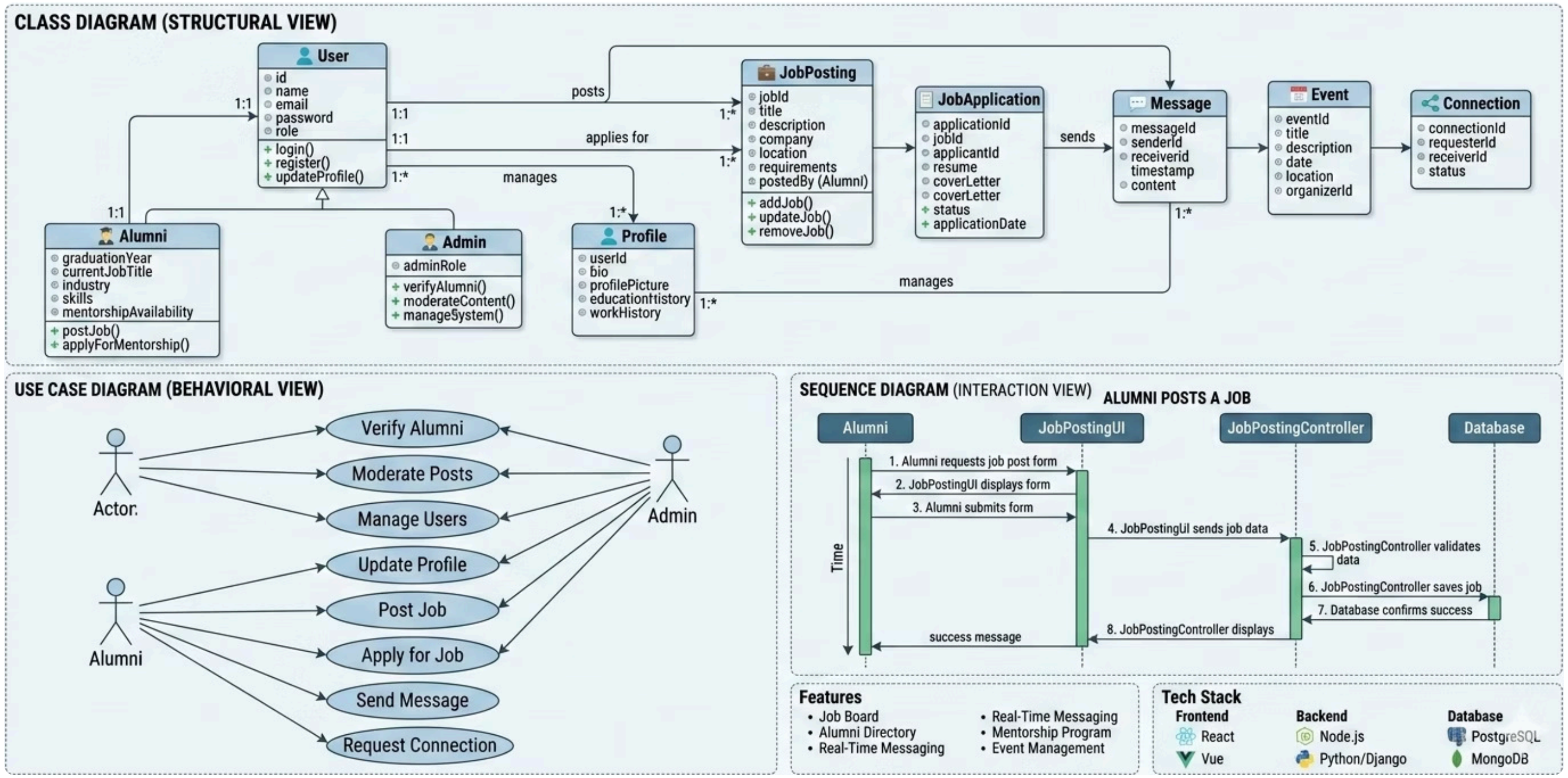
Directory & Search Module

Provides filterable talent grids and alumni directories, allowing users to search by department, location, and skills.

Data Flow Diagram (DFD) - Context Level



UML Diagram: System Use Cases



CONCLUSION

Project Conclusion & Future Vision

The Alumni Networking Portal provides a secure and scalable platform for alumni-student engagement.

Project Summary

- Centralized, secure alumni networking platform
- Enhances job sharing and mentorship opportunities
- Features AI-based personalization for users
- Implemented a comprehensive role-based authentication system

Future Scope

- Development of a dedicated mobile application
- Integration of real-time chat & video mentoring
- Advanced AI analytics for deeper insights
- Scalability to support multiple institutions

> Improves institutional placement support and alumni tracking

REFERENCES

References

A structured comparison of current platforms and research, highlighting deficiencies our project aims to address.

Platform / Research	Key Features	Identified Gaps
LinkedIn	Professional networking, job search, profile visibility	Not institution-specific, no alumni verification system
AlmaConnect	Alumni engagement platform, job postings, event management	Subscription-based model, limited AI integration
IEEE Alumni Management System	Centralized alumni database, structured communication	No AI-based personalization, limited job recommendation features
AI-Based Job Recommendation Systems	Resume-based job matching, machine learning models	Not integrated with alumni networking platforms

Thank You

Questions and Discussion.

Avanthi Institute of Engineering and Technology

Makavarapalem